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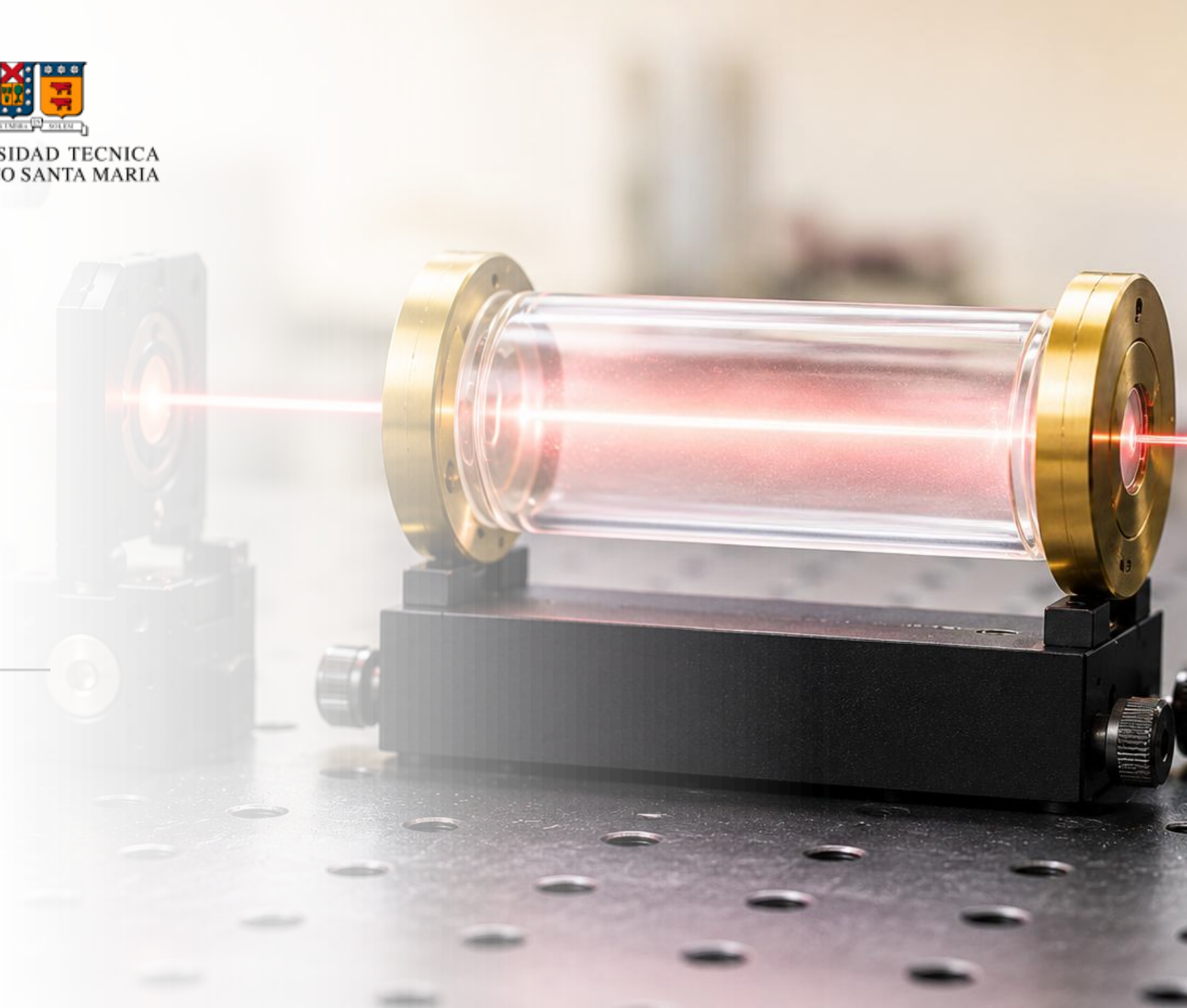
Universidad Tecnica Federico Santa Maria

Rubidium Absorption Spectrum

Alexis R. Legon

QSTATE
Quantum Systems Theory And Technology

QuAntü



Motivation

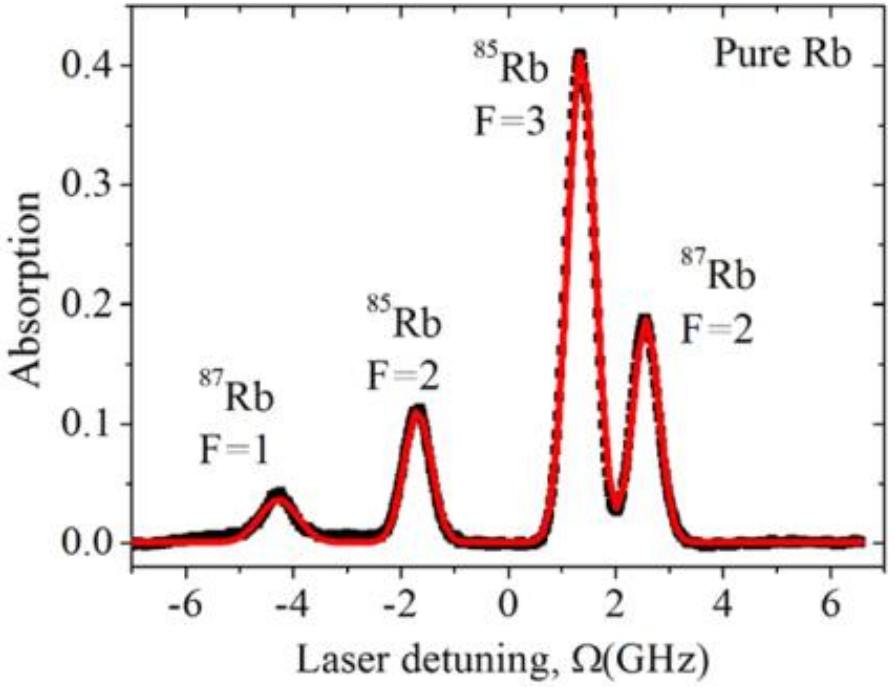
Reference Absorption Spectrum Used for Model Validation

Scientific questions

- What is the origin of the observed resonances?
- How does Doppler broadening affect the spectrum?
- What level of theory is required to reproduce the experiment?
- How does the description improve as more microscopic physics is included?

Goal

Reproduce and understand the absorption spectrum of a natural-abundance Rb vapor cell.



Ma et al, Scientific Reports 2017

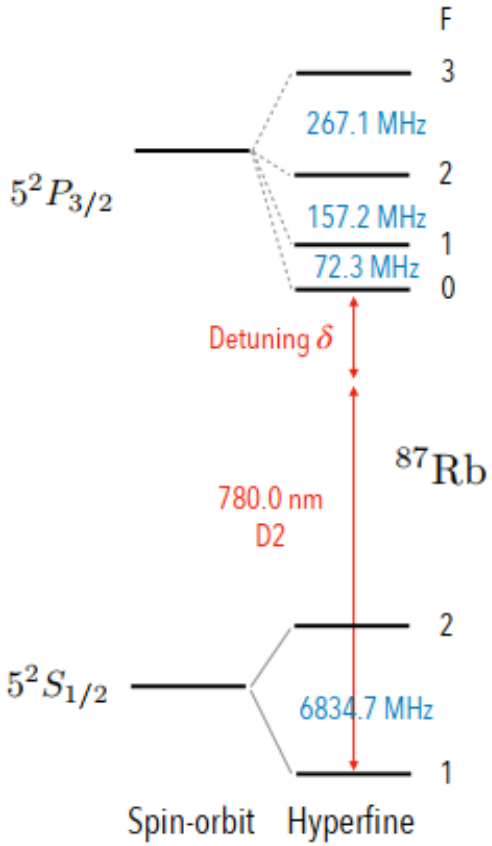
Temperature: **293 K**

Physical Origin of the Rubidium Absorption Spectrum

Natural-abundance Rb

$^{85}\mathrm{Rb}$ 72.17%

$^{87}\mathrm{Rb}$ 27.83%



Thermal motion

$$T = 293\,K$$

Maxwell-Boltzmann

$$f(v) = \sqrt{\frac{m}{2\pi k_B T}} e^{-mv^2/(2k_B T)}$$

Doppler broadening

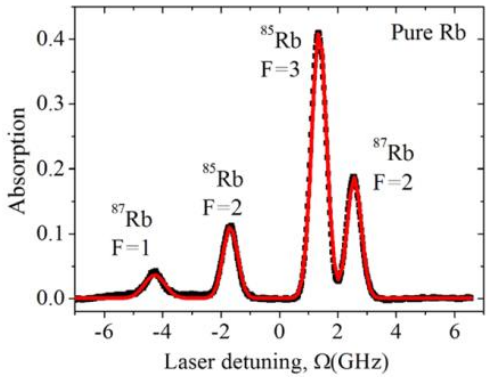
Doppler broadening dominates the observed absorption spectrum

Relevant frequency scales

$\Delta\nu_D \approx 500\,\mathrm{MHz}$ (Doppler width)

$\Gamma \approx 6\,\mathrm{MHz}$ (Natural linewidth)

Experimental data acquisition

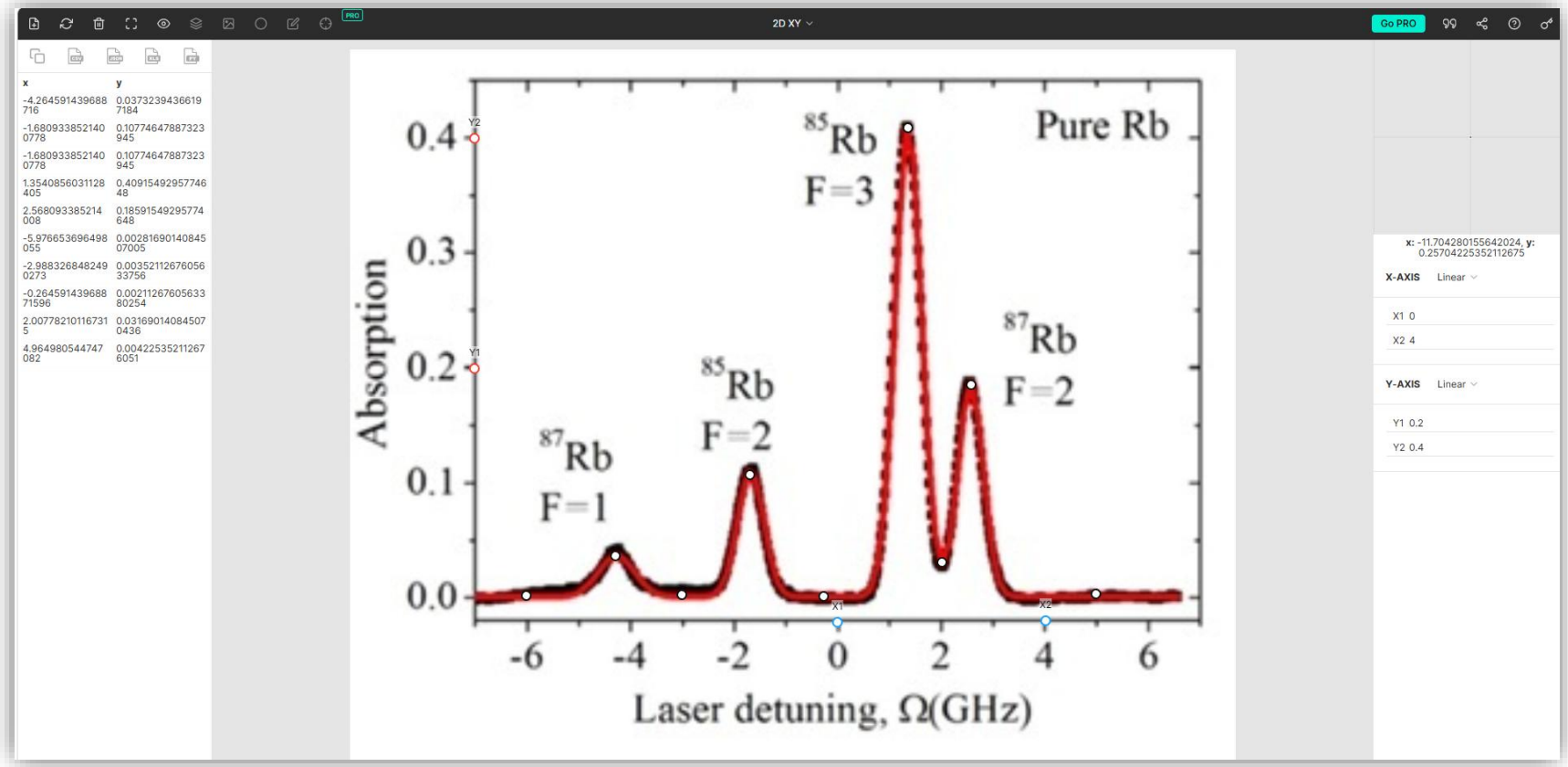


PlotDigitizer

Data

Normalization

Experimental dataset



Experimental data were reconstructed from the published absorption spectrum

Literature Review and Theoretical Approaches

Experimental Reference
Ma et al. (2017)
- Experimental absorption spectrum.
- Homogeneous and inhomogeneous broadening.
- Natural-abundance Rb vapor.
- Reference dataset for model validation.

Spectroscopic Techniques
Jacques et al. (2009)
- Doppler spectroscopy.
- Doppler-free spectroscopy.
- Saturated absorption spectroscopy.
- Hyperfine structure measurements.

Susceptibility-Based Models
Siddons et al. (2008)
- Electric susceptibility formalism.
- Absolute absorption spectroscopy.
- Weak-probe approximation

Susceptibility-Based Models
Agnew et al. (2024)
- Susceptibility-based absorption model.
- Maxwell–Boltzmann velocity distribution.
- Doppler and Voigt broadening.

Rate-Equation Models
Bala et al. (2022)
- Five-level rate-equation model.
- Hyperfine pumping.
- Transit relaxation.
- Doppler-broadened absorption.

Density-Matrix Approaches
Horn et al. (2011)
- Optical Bloch equations.
- Density matrix formalism.
- Atomic coherences and population dynamics.

The reviewed approaches motivate the development of increasingly realistic absorption models.

Modeling Strategy

Objective

Develop increasingly realistic theoretical models to reproduce the experimental absorption spectrum.

Experimental dataset

↓

Model 1
Phenomenological Gaussian model

↓

Model 2
Isotropic Doppler model

↓

Model 3
Microscopic susceptibility model

↓

Future work
Master-equation approach

Model	Main physical ingredient
Model 1	Doppler broadening
Model 2	Isotropic composition
Model 3	Optical susceptibility
Future work	Quantum dynamics

Increasing levels of physical realism are progressively incorporated.

Model 1: Phenomenological Gaussian Model

Physical idea

At room temperature, the dominant broadening mechanism is Doppler broadening.

Thermal motion

$$f(v) \propto e^{-mv^2/(2k_BT)}$$

$$T = 293\,K$$

↓

Doppler linewidth

$$\Delta\nu_D \approx 500\,\mathrm{MHz}$$

↓

Effective absorption model

↓

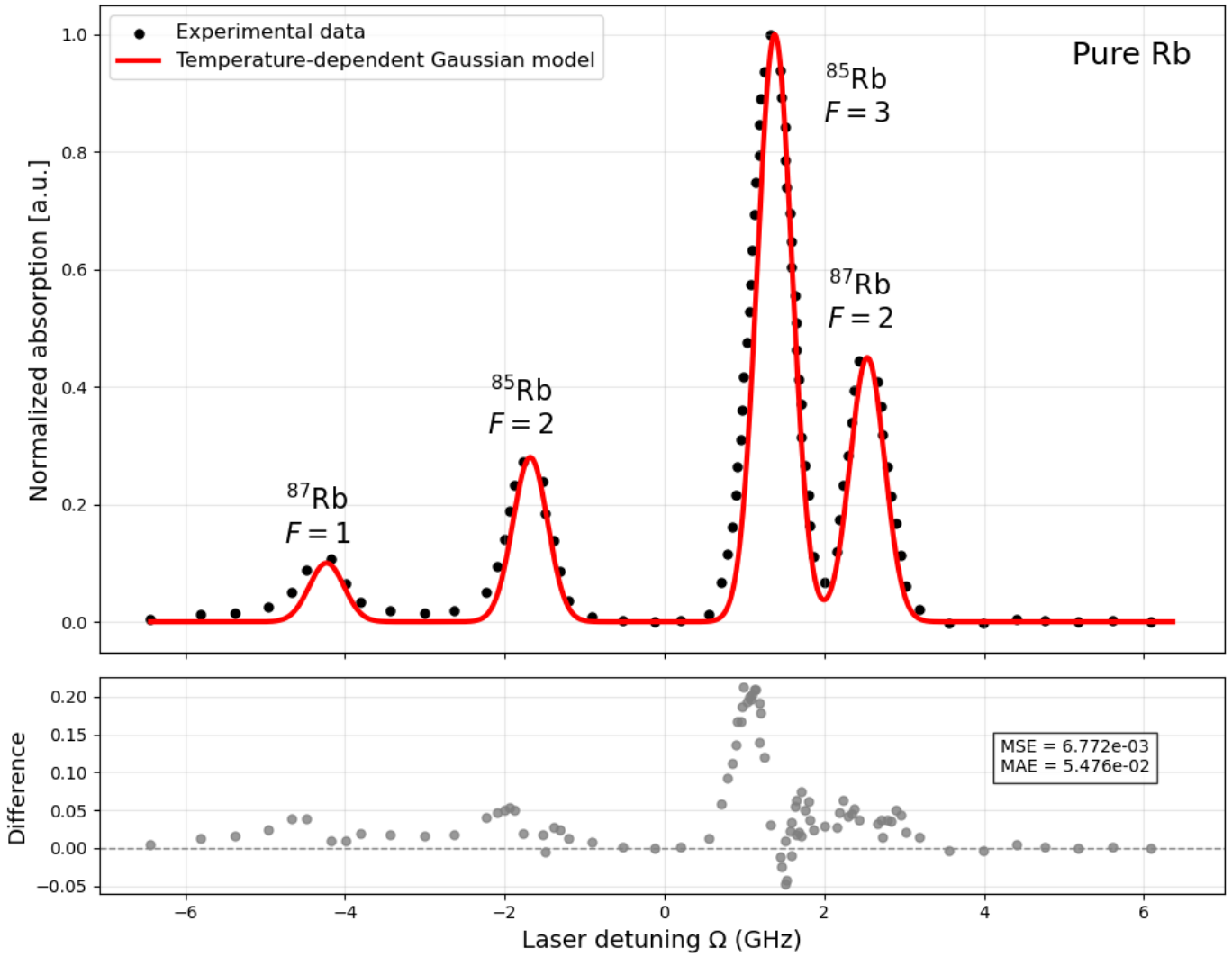
$$A(\Omega) = \sum_i A_i \exp\left[-\frac{(\Omega - \Omega_i)^2}{2\sigma^2}\right]$$

Model assumptions

- Four effective resonances.
- Common Doppler width.
- Phenomenological amplitudes.
- No isotropic structure.

Each absorption peak is approximated by an effective Doppler-broadened Gaussian profile.

Model 1 Results



Key observations

- Overall spectral structure is reproduced.
- Four main resonances are captured.
- Peak positions require phenomenological adjustments.
- No isotopic interpretation.

The Gaussian model reproduces the overall absorption profile but does not explicitly incorporate the isotopic structure of natural rubidium.

Model 2: Isotropic Doppler Model

Physical idea

The natural abundance of rubidium contains two isotopes that contribute differently to the absorption spectrum.

Natural-abundance Rb

$^{85}\mathrm{Rb}\quad 72.17\%\qquad ^{87}\mathrm{Rb}\quad 27.83\%$

↓

Hyperfine resonances

$^{87}\mathrm{Rb} : F = 1,\ F = 2$

$^{85}\mathrm{Rb} : F = 2,\ F = 3$

↓

Isotopic absorption spectra $A_{85}(\Omega)$ $A_{87}(\Omega)$

↓

Total absorption model

↓

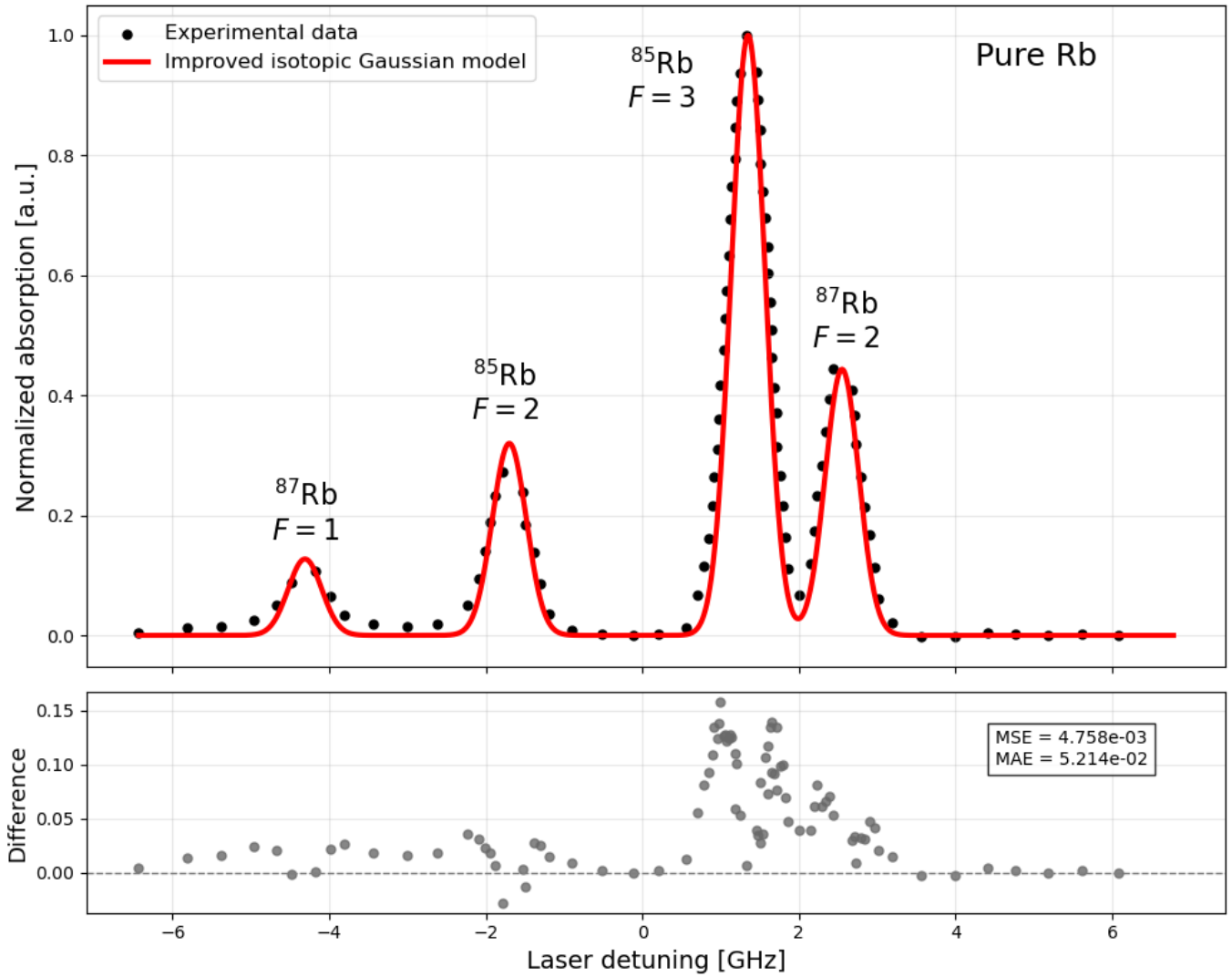
$$A(\Omega) = A_{85}(\Omega) + A_{87}(\Omega)$$

Model assumptions

- Natural isotropic abundances.
- Effective hyperfine resonances.
- Independent isotopic contributions.
- Doppler broadening retained.

The isotopic composition of natural rubidium is explicitly incorporated into the absorption spectrum.

Model 2 Results



Key observations

- Overall spectral structure is reproduced.
- Isotopic contributions are explicitly included.
- Improved peak positions and relative amplitudes.
- Physical interpretation of the four resonances.

Including the natural isotopic composition improves the agreement with the experimental spectrum and provides a physical interpretation of the observed resonances.

Model 3: Microscopic Susceptibility Model

Physical idea

The absorption spectrum is determined by the optical response of the atomic medium.

Optical susceptibility

$$\chi(\omega) = \sum_i \frac{A_i}{\omega - \omega_i + i\Gamma_i}$$

↓

Absorption

$$A(\omega) \propto \mathrm{Im}\left[\chi(\omega)\right]$$

↓

Doppler broadening

$$\omega' = \omega - kv$$

↓

Velocity averaging

↓

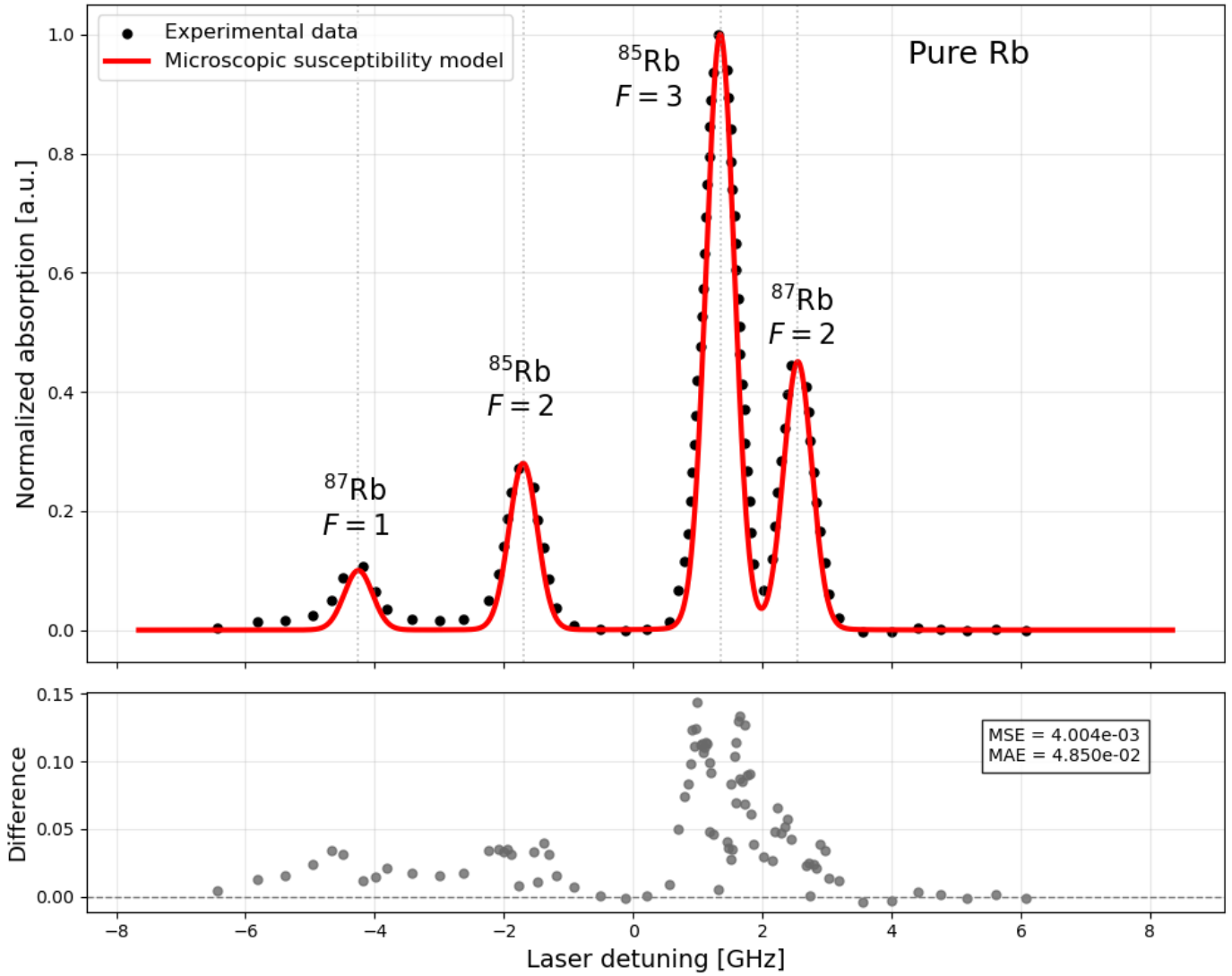
$$\chi_D(\omega) = \int dv\, f(v)\, \chi(\omega, v)$$

Model assumptions

- Optical susceptibility formalism.
- Maxwell–Boltzmann velocity distribution.
- Homogeneous broadening included.
- Doppler broadening included.

The absorption spectrum is obtained from the Doppler-averaged optical susceptibility.

Model 3 Results



Key observations

- Best agreement with experiment.
- Homogeneous and Doppler broadening included.
- Microscopic optical response.
- Lowest MSE and MAE.

The susceptibility-based model provides the most accurate description of the experimental absorption spectrum.

Next Step: Siddons Framework

Physical idea

Towards a quantitative description of the rubidium absorption coefficient.

Hyperfine transitions

$$F \rightarrow F'$$

↓

Optical susceptibility

$$\chi(\omega)$$

↓

Atomic density

$$N(T)$$

↓

Absorption coefficient

↓

$$\alpha(\omega) \propto N(T)\,\mathrm{Im}[\chi(\omega)]$$

Additional physical ingredients

- Hyperfine transition strengths.
- Temperature-dependent atomic density.
- Real spectroscopic parameters.
- Absolute absorption coefficient.
- Quantitative comparison with experiment.

The absorption coefficient is obtained from microscopic atomic and spectroscopic parameters.

From Absorption Models to Atomic Dynamics

Physical idea

The absorption spectrum originates from the microscopic dynamics of atomic populations and coherences.

Atomic populations

ρ_{ii}

↓

Atomic coherences

ρ_{ij}

↓

Density matrix description

ρ ↓

Optical susceptibility

↓ $\chi(\omega)$

Absorption spectrum

↓

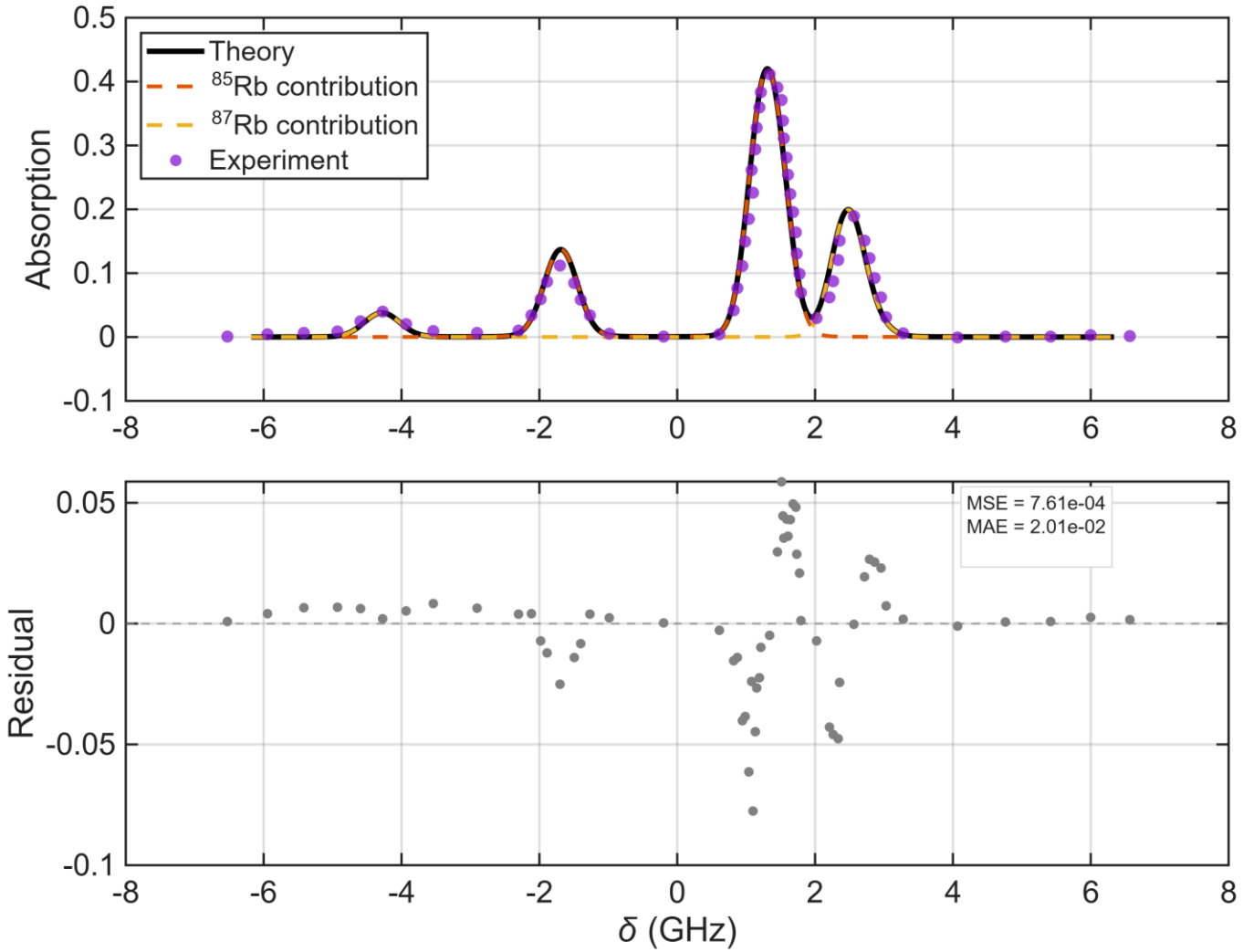
$A(\omega)$

Additional physical ingredients

- Optical pumping.
- Relaxation processes.
- Population redistribution.
- Atomic coherences.
- Microscopic light–matter interaction.

The absorption spectrum emerges from the dynamics of atomic populations and coherences.

Density-Matrix Results



Microscopic population and coherence dynamics reproduce the observed absorption spectrum.

Conclusions

Main results

- Doppler broadening dominates the room-temperature spectrum.
- The isotopic composition of natural rubidium is essential to reproduce the observed resonances.
- Optical susceptibility provides a microscopic connection between atomic transitions and absorption.
- Density-matrix calculations yield the best agreement with the experimental spectrum.

Outlook

- Implementation of the complete Siddons formalism.
- Extension toward fully predictive microscopic spectroscopic models.

The Rb absorption spectrum can be understood through the progressive incorporation of microscopic physical mechanisms.

References

- Siddons et al., Journal of Physics B (2008)
- Jacques et al., European Journal of Physics (2009)
- Horn et al., Physical Review A (2011)
- Ma et al., Scientific Reports (2017)
- Bala et al., Journal of Physics B (2022)
- Agnew et al., AIP Conference Proceedings (2024)